Gap Analysis — Offline Bangla Dubbing Video Player

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Summary

Updated for BRD version 1.1: 14 of the 18 business requirements are High feasibility, 4 are Medium, and none are Low. The spike's licence check and the rework plan's fallbacks removed the two biggest gaps from version 1.0. Three gaps still carry real risk:

1. **Bangladeshi accent (BR-04b), Low → Medium:** SUBAK.KO, a 241-hour Bangladeshi Bangla corpus under CC-BY-4.0, makes fine-tuning possible. It is now a Should for phase 2, so it no longer blocks version 1.
2. **Speaker voices (BR-05), Medium:** diarization accuracy is unproven. Scope is now 4 guaranteed speakers, and BR-17 lets users fix mistakes.
3. **Live mode (BR-09), Medium:** still depends on GPU speed, but it is no longer a release blocker.

The licence gap (NFR-08) is **closed**: a commercial-safe stack was verified in the spike.

This analysis is based on the BRD version 1.1 dated 2026-10-29 and the spike's licence check. Speed, quality and accuracy remain unproven until the spike prototype runs on real hardware.

Current vs target state

Aspect	Current state	Target state
How users watch foreign video	Bangla subtitles, or not at all	Spoken Bangladeshi Bangla, no reading needed
Dubbed content available	Only what studios or TV channels dub	Any local video file
Existing AI dubbing	Online, paid per minute, uploads private video	Offline, no fees, nothing leaves the device

Aspect	Current state	Target state
Multiple speakers	Single generic voice in most tools	A distinct, consistent voice per speaker
Background audio	Often lost or muffled	Original music and effects kept
Platforms	Mostly web or one OS	Windows, macOS and Linux desktop

Gap per business requirement

Feasibility: **High** = ready tools exist, **Medium** = tools exist but need tuning, **Low** = needs new data or research.

ID	Available today	Gap	Action to close	Feasibility
BR-01 Player	PyQt6 + python-mpv (spike player built)	None	Turn spike player into app module	High
BR-02 Language detect	faster-whisper auto-detect; Whisper translate-to-English	Non-English, non-Hindi quality unproven	English and Hindi guaranteed; others best-effort via pivot	High
BR-03 Speech replaced	Spike pipeline written end to end	Not yet run on models	Run spike (E1)	High
BR-04a Bangladeshi vocabulary	IndicTrans2 (MIT)	Output not dialect-aware	Build a Bangladeshi glossary; optional LLM post-edit	Medium
BR-04b Bangladeshi accent	Indic Parler-TTS presets; SUBAK.KO corpus (CC-BY-4.0)	Voices lean Indian Bengali; corpus needs cleaning for TTS	Phase 2: fine-tune TTS on SUBAK.KO	Medium

ID	Available today	Gap	Action to close	Feasibility
BR-05 Voice per speaker	pyannote 3.1 (MIT); 6 Bengali voices	Accuracy unproven; overlap errors	Scope to 4 guaranteed speakers; BR-17 for fixes	Medium
BR-06 Keep background	Demucs	Artefacts on dense music	Fallback: lower original audio under dub	High
BR-07 Fully offline	All models run locally; spike has `--offline` mode	First-run download	Model manager with resume and checksums	High
BR-08 Pre-process mode	Spike pipeline	Long Bangla lines	Time-stretch capped at 1.35×; NFR-05 tolerance	High
BR-09 Live mode	Same components	Several seconds of latency per line	Rolling buffer, GPU only; not a release blocker	Medium
BR-10 Choose voices	TTS presets	No UI	Speaker-voice mapping panel	High
BR-11 Switch audio	Spike player does this	None	Port to app	High
BR-12 Export	Spike writes two-track MKV (tested)	None	Export dialog	High
BR-13 Subtitles	Spike writes Bangla SRT (tested)	None	Show in player	High
BR-14 Cache	Local storage	None	Cache keyed by file hash	High
BR-15 Edit lines	TTS can regenerate single lines	No editor UI	Line editor panel	High
BR-16 Bangla UI	Qt translation system	Translation work	Qt Linguist strings	High

ID	Available today	Gap	Action to close	Feasibility
BR-17 Fix speakers	Diarization output per line	No UI	Rename, merge, split and re-dub lines	High

Technology and licence gaps

Licences were verified against model cards and repositories during the spike on 2026-10-29 (sources in the spike document).

Component	Candidate	Licence (verified)	Commercial use	Status in BRD v1.1
Speech-to-text	faster-whisper (Whisper large-v3 / turbo)	MIT	Yes	Chosen
Diarization	pyannote speaker-diarization-3.1	MIT (gated download)	Yes	Chosen; Hugging Face token needed once
Separation	Demucs	MIT	Yes	Chosen
Translation	IndicTrans2	MIT	Yes	Chosen; English ↔ Indic only
Translation	NLLB-200	CC-BY-NC 4.0	No	Excluded (NFR-08)
Translation	Local LLM via llama.cpp	Varies by model	Varies	Alternative if E2 fails; check each model's licence
Bangla TTS	Indic Parler-TTS	Apache 2.0 (gated download)	Yes	Chosen; 6 Bengali voices
Bangla TTS	MMS-TTS (ben)	CC-BY-NC 4.0	No	Excluded (NFR-08)
Voice data	SUBAK.KO	CC-BY-4.0	Yes, with attribution	Phase 2 fine-tuning for BR-04b

Component	Candidate	Licence (verified)	Commercial use	Status in BRD v1.1
Voice cloning	XTTS-v2	Coqui Public Model Licence	No	Out of scope
Player	PyQt6 / python-mpv	GPL or commercial (Qt); mpv LGPL/GPL	Conditional	PySide6 (LGPL) if the app is closed-source

Chosen stack for the commercial product: faster-whisper, pyannote 3.1, Demucs, IndicTrans2, Indic Parler-TTS, with PySide6 or PyQt6 under a commercial licence. One open item: decide PySide6 vs commercial PyQt6 before development starts.

Requirement analysis

Refined requirements

ID	BRD v1.1 wording	Testable acceptance
BR-03	Replace speech with spoken Bangla	Every detected speech segment of 1 s or more has a Bangla audio line; untranslated segments fall back to original audio
BR-04a	Bangladeshi vocabulary	Glossary terms used in 95%+ of matching lines; reviewers rate wording 3.5/5 or higher
BR-04b	Bangladeshi accent (phase 2)	70%+ of Bangladeshi UAT testers rate the voice as "sounds Bangladeshi"
BR-05	Voice per speaker, 4 guaranteed	90%+ correct speaker labels with up to 4 speakers; same speaker keeps the same voice across the file
BR-06	Keep background, with fallback	Reviewers confirm music is audible; fallback triggers automatically when separation fails
BR-09	Live mode on recommended hardware	Playback starts within 60 s and does not pause for buffering on a 2-hour film; disabled below recommended hardware

ID	BRD v1.1 wording	Testable acceptance
BR-17	Fix speakers	A merge or split re-dubs only the affected lines, in under 1 min for a 2-hour film on recommended hardware

Dependencies (build order)

1. Audio extraction (ffmpeg) is needed by everything else.
2. Speech-to-text feeds translation; diarization feeds voice assignment.
3. Translation feeds TTS; TTS feeds timing fit; timing fit and separation feed the mix.
4. Pre-process mode (BR-08) must work before live mode (BR-09).
5. Voice selection, line editing and export build on the finished pre-process pipeline.

Working assumptions for the BRD's open questions

Most BRD open questions were decided in version 1.1; three remain open.

Question	Status in BRD v1.1
Commercial, free or internal?	Decided: commercial
First source languages	Decided: English and Hindi guaranteed; others best-effort
Bangladeshi voice data	Decided: SUBAK.KO for phase 2
Buffer and preset voices acceptable?	Decided: yes for version 1
Minimum hardware	Open; working assumption: CPU for pre-process, 8 GB GPU or Apple Silicon 16 GB for live
Budget and release date	Open; needed before development
Team names	Open; needed before development

Decisions needed and spike goals

Decisions for the product owner

- [x] ~~Commercial or non-commercial release: **commercial** (BRD v1.1)~~
- [x] ~~Accept Indian-Bengali preset voices for version 1: **yes**; Bangladeshi accent in phase 2~~

- [ ] Confirm minimum supported hardware
- [ ] Confirm the proposed success targets in the BRD
- [ ] Choose PySide6 or commercial PyQt6

What step 3 (spike / feasibility) must prove

#	Question	Pass condition
1	Does the end-to-end chain work offline?	A 5-minute English clip dubs into Bangla with no network
2	Which translation sounds most natural?	IndicTrans2 vs NLLB vs local LLM, scored by 3+ Bangladeshi reviewers
3	How Bangladeshi do the TTS voices sound?	Reviewer score recorded for each candidate voice
4	Does diarization hold up?	90%+ correct speaker labels on a 3-speaker clip
5	How fast is it?	Processing time per minute of video measured on GPU, Apple Silicon and CPU
6	Does timing fit work?	Dubbed lines start within 0.5 s of the original

Under BRD v1.1, only an E1 failure (no working offline chain) sends the project back to step 1. Failures in E3–E8 now reduce scope through the fallbacks already in the BRD, as set out in the rework plan.

Change log

Version	Date	Change
1.1	2026-10-29	Aligned with BRD v1.1: BR-04 split, BR-17 added, BR-05/06/09 rescoped; feasibility now 14 High, 4 Medium, 0 Low
1.1	2026-10-29	Licences verified in the spike; NLLB-200 and MMS-TTS excluded; licence gap closed
1.1	2026-10-29	Refined acceptance conditions and open questions updated to BRD v1.1 decisions

Version	Date	Change
1.0	2026-10-29	First version